

YOLO Detection Dataset Spectral Comparison - common 7-80 kHz analysis bandpass

Particles are exact YOLO label intervals.

Dataset noise is made of FFT windows that do not overlap any label.

Standalone Noise is a single external reference, shown once in neutral color.

All datasets and the standalone Noise reference are analyzed after the same 7-80 kHz bandpass filter.

Relative percentages are comparable within the retained passband; absolute target-band energy is reported separately in dB.

Datasets:

- YOLO v3 old pipeline: P1/yolo_dataset_v3
- P0 C1 Hamming: P0/data/dataset_hamming_c1_yolo
- P0 C1 clean + 7-80 kHz: P0/data/dataset_hamming_c1_clean_filt_yolo_trainval
- Standalone Noise reference: P0/data/Noise

Coverage summary:

P0 C1 clean + 7-80 kHz test: signals=578, labels=895, mean coverage=15.7%, p95=36.6%

P0 C1 Hamming test: signals=578, labels=6905, mean coverage=66.6%, p95=93.9%

YOLO v3 old pipeline test: signals=354, labels=384, mean coverage=7.0%, p95=14.1%

10-40 kHz and overlap summary:

YOLO v3 old pipeline 2um: particle=88.5%, dataset-noise=64.0%, Noise reference=59.8%, dominant overlap=True

YOLO v3 old pipeline 4um: particle=96.8%, dataset-noise=64.0%, Noise reference=59.8%, dominant overlap=True

YOLO v3 old pipeline 10um: particle=87.9%, dataset-noise=64.0%, Noise reference=59.8%, dominant overlap=True

P0 C1 Hamming 2um: particle=66.6%, dataset-noise=60.5%, Noise reference=59.8%, dominant overlap=True

P0 C1 Hamming 4um: particle=77.3%, dataset-noise=60.6%, Noise reference=59.8%, dominant overlap=True

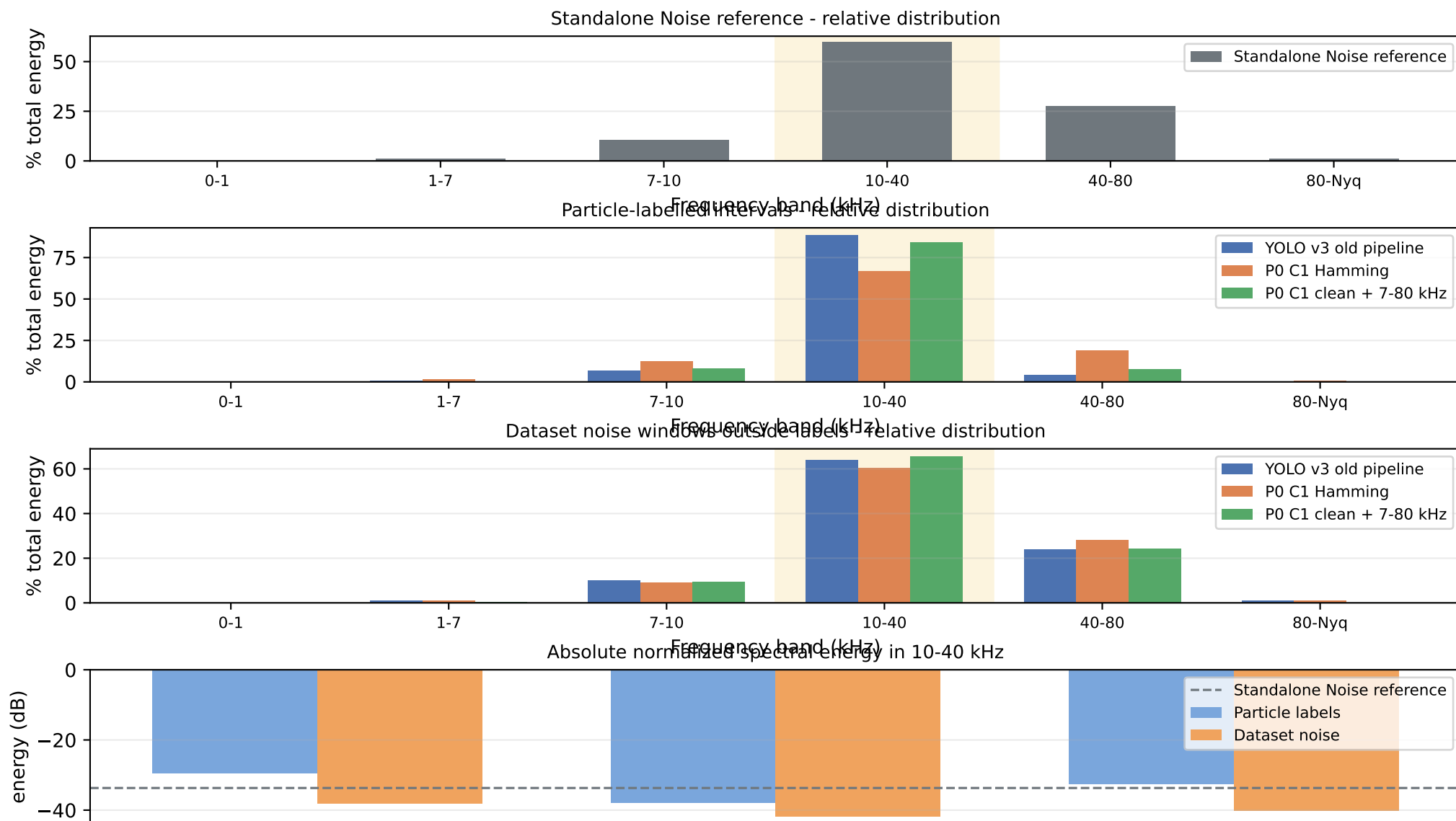
P0 C1 Hamming 10um: particle=67.0%, dataset-noise=60.6%, Noise reference=59.8%, dominant overlap=True

P0 C1 clean + 7-80 kHz 2um: particle=84.0%, dataset-noise=65.7%, Noise reference=59.8%, dominant overlap=True

P0 C1 clean + 7-80 kHz 4um: particle=96.9%, dataset-noise=65.7%, Noise reference=59.8%, dominant overlap=True

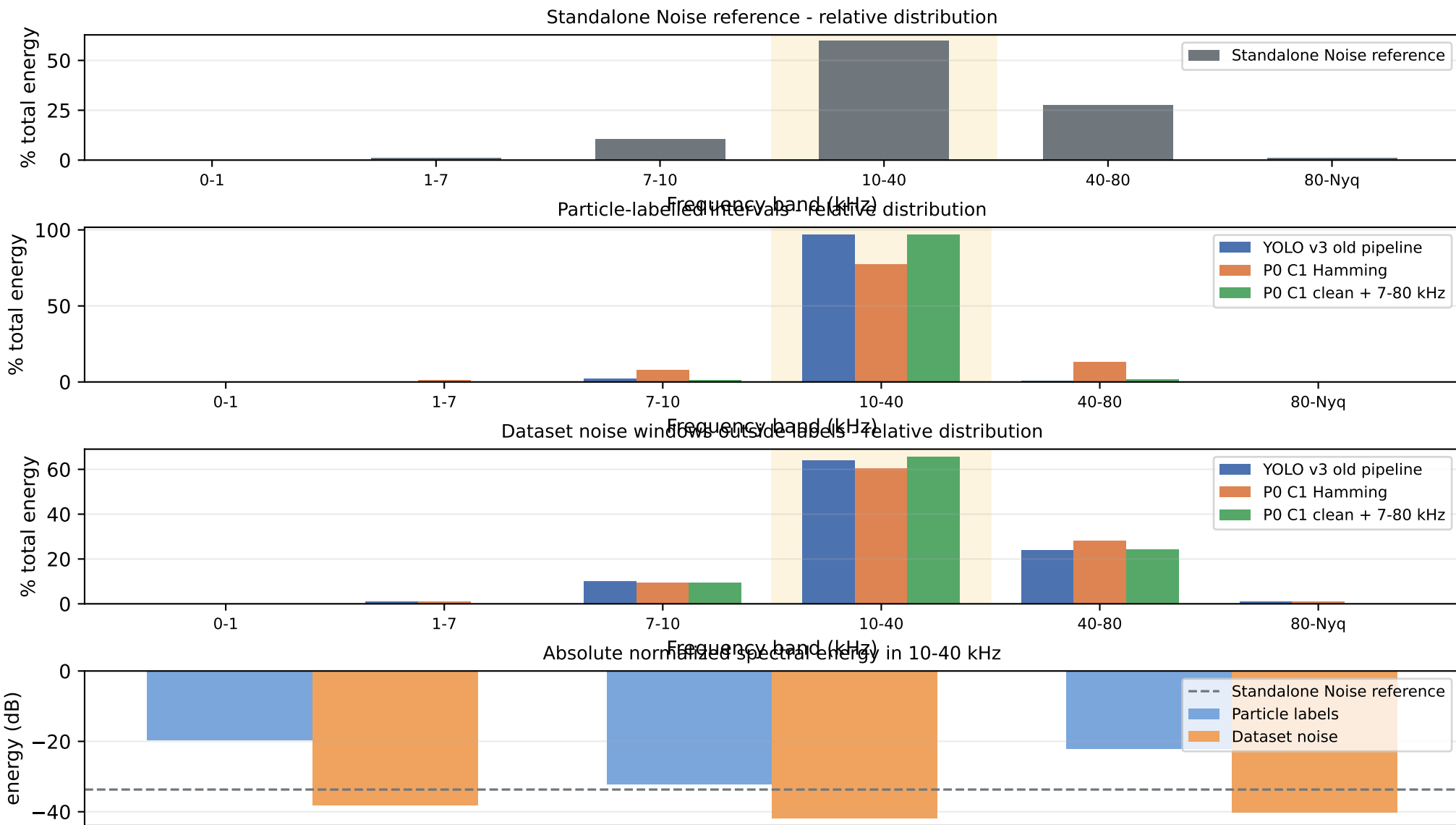
P0 C1 clean + 7-80 kHz 10um: particle=82.0%, dataset-noise=65.7%, Noise reference=59.8%, dominant overlap=True

2um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



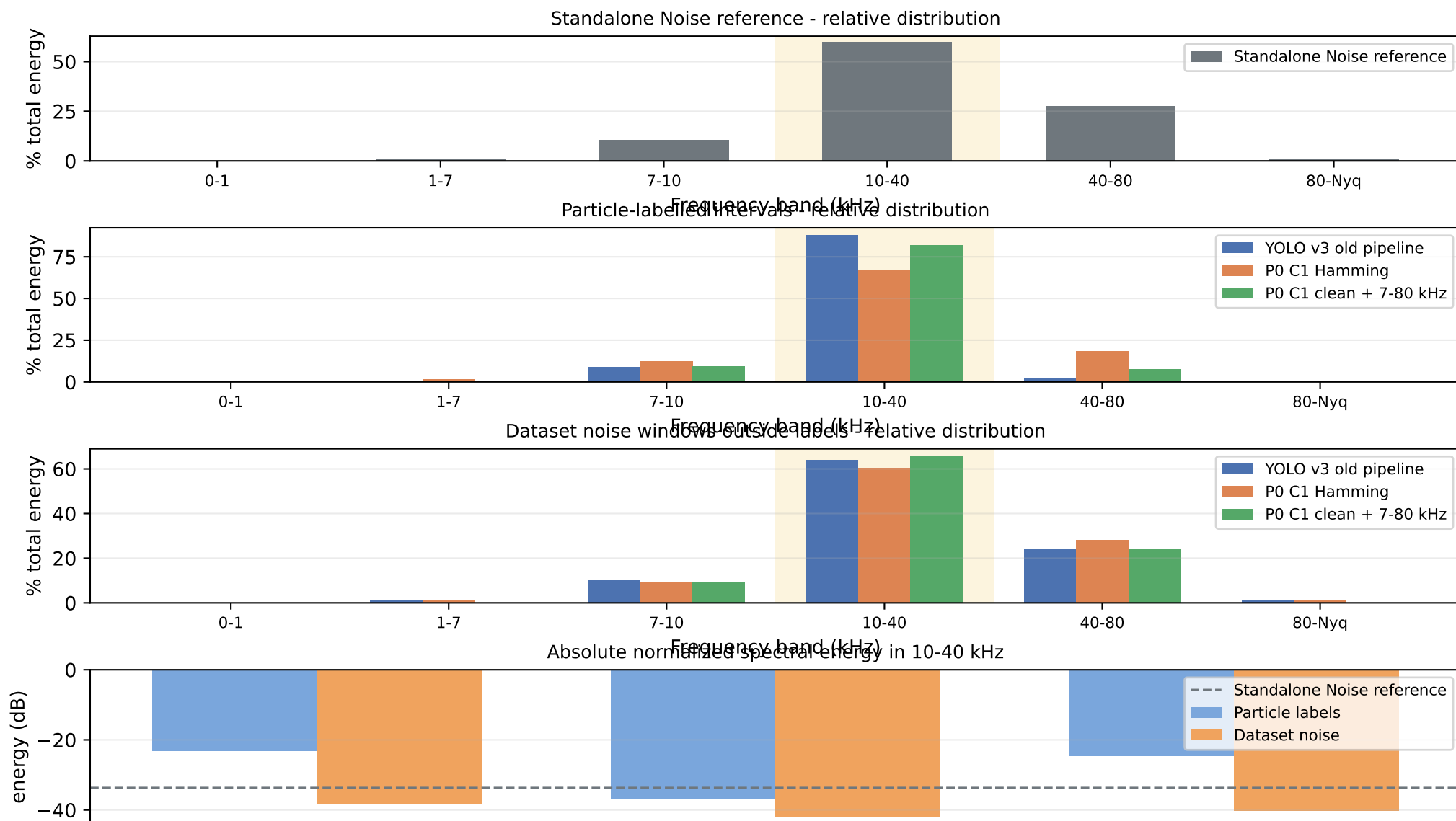
Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	88.5%	64.0%	59.8%	7.0%
P0 C1 Hamming	66.6%	60.5%	59.8%	66.6%
P0 C1 clean + 7-80 kHz	84.0%	65.7%	59.8%	15.7%

4um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	96.8%	64.0%	59.8%	7.0%
P0 C1 Hamming	77.3%	60.6%	59.8%	66.6%
P0 C1 clean + 7-80 kHz	96.9%	65.7%	59.8%	15.7%

10um - YOLO v3 old pipeline vs P0 C1 Hamming vs P0 C1 clean + 7-80 kHz



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 old pipeline	87.9%	64.0%	59.8%	7.0%
P0 C1 Hamming	67.0%	60.6%	59.8%	66.6%
P0 C1 clean + 7-80 kHz	82.0%	65.7%	59.8%	15.7%